The Wall-Tap Delay, Locked Structurally

Mediation, Eikonal Origin, the Fold Orbifold, Timing Eigenstructure, and the Rhythm Transition as Theorems

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Abstract

The wall-tap papers established their claims with synthetic batteries. This paper turns the claims that can be made structural into theorems, and states exactly which remain room constants or empirical values.

Three delays. Influence delay, identification delay and reported latency are distinct.

Mediation. Delays form a weighted reachability structure: a Lawvere metric, asymmetric under wind. Before the mediated arrival $d(s, r)$, intervention laws coincide and the optimal equal-prior error is exactly $\frac{1}{2}$. Afterwards it is nonincreasing along nested histories. Confirmation time $t_\varepsilon \geq d(s, r)$, possibly strictly or infinitely.

Origin. With a hyperbolic medium, first arrivals are geodesic distances of the acoustic metric. In wind this metric is of Randers type, with round trip $2Dc/(c^2 - w^2)$. Walls act by image sources. Positivity in seconds needs a finite speed and a counted clock; mediation alone is dilation-null.

Topology. Exact onset is not continuous even in total variation. Threshold times are continuous exactly under strict crossing. Observable onset and t_ε descend to the observational quotient; latent latency decompositions do not.

Snap and split. An overlap channel equivariant under $b \mapsto 2d - b$ makes the fold an orbifold point. There the normal derivative vanishes, Fisher rank drops, and the identification modulus is $\varepsilon^{1/2}$. An anti-invariant (signed) channel splits it.

Eigenstructure.

- Every symmetry of the record law gives an exact Fisher null vector, its generator: dilation (x, c, L) and gain–contrast $(r, -g)$. Anchors remove exactly these.
- Wall eigenvalues scale as ε^2.
- Timing precision is Woodward's $E\beta^2/\sigma^2$.

The rhythm transition. For a room with an exponential diffuse tail,

$$T_c = \frac{\tau_d}{2} \ln\Big[\frac{\theta}{1-\theta}\,\frac{a^2\tau_d}{2e_0\tau_p}\Big],$$

and in general $T_c[h(\cdot/k)] = k\,T_c[h]$. The 35.58 ms of the battery is a room constant, not a universal one.

Evidence. Every statement is proved, with exact or symbolic witnesses and no random sampling.

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1 Three delays

Definition 1.1 (Time-indexed experiment)**.** For a source intervention A and the receiver record $R_{\le t}$, set $\mathcal{E}_t = \{\mathcal{L}(R_{\le t} \mid \mathrm{do}(A = a)) : a \in \mathcal{A}\}$. Then:

- the *influence delay* is $\tau_{\mathrm{obs}} = \inf\{t : \mathcal{E}_t \text{ is not constant in } a\}$;
- the *identification delay* is $t_\varepsilon = \inf\{t : r^*(t) \le \varepsilon\}$ for a declared task;
- the *reported latency* is the apparatus interval between trigger and detection convention, the composite $\tau_{\mathrm{trigger}} + \tau_{\mathrm{source}} + \tau_{\mathrm{path}} + \tau_{\mathrm{transducer}} + \tau_{\mathrm{detector}}$ of the original paper.

A response may have arrived while still too weak to identify. Conversely, an estimator can locate a pulse centre precisely without any strict wavefront having been identified: a Gaussian pulse has noncompact support.

2 Mediation: no distinguishability before arrival

Definition 2.1 (Causal delay network)**.** A finite directed acyclic graph of source, mediator and receiver nodes, with these properties:

- the source acts at time 0;
- the background randomness U is independent of the intervention;
- each node is a causal function of its incoming records and of U;
- each edge e has a lower transmission time $\delta_e \ge 0$;
- the receiver has no undeclared source-dependent input.

Set $d(s, r) = \inf_{\gamma : s \rightsquigarrow r} \sum_{e \in \gamma} \delta_e$, with $d = \infty$ if there is no path.

Theorem 2.2 (Mediation). *For $t < d(s,r)$, $\mathcal{L}(R_{\le t} \mid \mathrm{do}(A = a)) = \mathcal{L}(R_{\le t} \mid \mathrm{do}(A = a'))$ for all a, a'.*

Proof. Couple the two interventions on the same U. Before time 0 the histories agree. A difference at a node at time t needs a difference on some incoming edge at time $\le t - \delta_e$. Induction along the acyclic order shows that receiver histories agree pathwise before $d(s,r)$, so the laws agree. □

Proposition 2.3 (The delay structure is a Lawvere metric). *d satisfies $d(s,s) = 0$ and $d(s,u) \le d(s,r) + d(r,u)$. It may be asymmetric, and it may take the value ∞. The network is therefore a category enriched over $([0,\infty], \ge, +)$, a Lawvere metric space [1]. Rescaling every δ_e by $k > 0$ is an automorphism of the structure.* Mediation fixes order and composition, not seconds.

Theorem 2.4 (Testing and confirmation). *For any decision based on $R_{\le t}$, the error probabilities satisfy $\alpha_t + \beta_t \ge 1 - \mathrm{TV}(P_0^t, P_1^t)$. The optimal equal-prior error is*

$$r^*(t) = \tfrac{1}{2}\big(1 - \mathrm{TV}(P_0^t, P_1^t)\big).$$

Hence:

(a) for $t < d(s,r)$, $r^*(t) = \frac{1}{2}$;

(b) r^* is nonincreasing in t, because restricting a later record to an earlier one is a garbling;

(c) $t_\varepsilon \ge d(s,r)$;

(d) for any adapted alarm time T with false-alarm probability at most α, $P_1(T \le t) \le \alpha + \mathrm{TV}(P_1^t, P_0^t)$.

Proof. For a test $\varphi \in [0,1]$, $\alpha + \beta = 1 - (\mathbb{E}_1\varphi - \mathbb{E}_0\varphi)$, and the supremum of the difference is the total variation, attained by a likelihood-ratio test. For (b), apply Corollary 3.5 of *Experiments, Simulation and Descent*. For (d), the event $\{T \le t\}$ is $R_{\le t}$-measurable. □

Witness (X1, X2). A network with a direct path of length 3 and a wall path of $2 + 2$, with receiver noise $\frac{1}{5}$:

- $\mathrm{TV} = 0, 0, 0, \frac{3}{5}, \frac{3}{5}, \frac{99}{125}, \frac{99}{125}$ for $t = 0, \dots, 6$;
- $r^* = \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{5}, \frac{1}{5}, \frac{13}{125}, \frac{13}{125}$.

Remark 2.5 (Mediation is not garbling without the full cut). The receiver cannot outperform a mediator only if $A \to M_{\le t} \to R_{\le t}$ is a statistical chain. Correlated side information at the receiver must be included in the cut.

3 The geometric origin of the delay

Imported Theorem 3.1 (Finite propagation and first arrival [2, 3]). *For a hyperbolic equation, i.e. one with real characteristic roots (the wall-tap acoustic lock, WT13), solutions have finite propagation speed. The first-arrival time between two points is the distance in the metric of the characteristic cone, by Fermat's principle and the eikonal equation. For a stationary moving medium this distance is of Randers type [4].*

Corollary 3.2 (Delay as geodesic distance). *(a) In still air with speed c: $t_{\mathrm{dir}} = x/c$. By the image source, $t_{\mathrm{echo}} = (2L - x)/c$, so $t_{\mathrm{echo}} - t_{\mathrm{dir}} = (2L - 2x)/c = d(x)$.*

(b) In uniform wind w along the path, the round trip is $D/(c+w) + D/(c-w) = 2Dc/(c^2 - w^2)$. The echo distance is $D/(1 - w^2/c^2)$ (WT11). The asymmetry is the Randers one-form, and a reciprocity channel recovers w.

(c) *The mediation weights δ_e of Theorem 2.2 are these geodesic times. Positivity in seconds requires $c < \infty$, $D > 0$ and a counted clock. The triple $(x, c, L) \mapsto (kx, kc, kL)$ leaves every delay invariant (X4, X6).*

No-Go 3.3 (Causality alone gives no dead interval). A causal response kernel may begin immediately after excitation. A strictly positive onset needs a speed bound and a positive distance, both of which are declarations about the medium.

4 Topology of delay

Proposition 4.1 (Exact onset is fragile). *$\mu_n = (1 - \frac{1}{n})\delta_T + \frac{1}{n}\delta_0$ satisfies $\mathrm{TV}(\mu_n, \delta_T) = \frac{1}{n} \to 0$. Yet every μ_n has onset 0, while the limit has onset T (X3). Exact onset is not continuous in total variation, nor in any weaker topology.*

Theorem 4.2 (Stability of confirmation times). *Let $r_k^* \to r^*$ uniformly on $[0, t_{\max}]$, with each function nonincreasing. Suppose r^** crosses strictly *at t_ε: $r^*(t) > \varepsilon$ for $t < t_\varepsilon$, and $r^*(t) < \varepsilon$ for $t > t_\varepsilon$. Then $t_{\varepsilon,k} \to t_\varepsilon$. Without strict crossing the conclusion fails.*

Proof. Fix $\eta > 0$. By strictness and monotonicity, $r^*(t_\varepsilon - \eta) > \varepsilon + \delta$ and $r^*(t_\varepsilon + \eta) < \varepsilon - \delta$ for some $\delta > 0$. Uniform convergence within δ places $t_{\varepsilon,k}$ in $[t_\varepsilon - \eta, t_\varepsilon + \eta]$.

Counterexample. If $r^* \equiv \varepsilon$ on $[1, 2]$, the perturbations $\pm 1/(10k)$ make $t_{\varepsilon,k}$ alternate between 1 and 2 (X3). □

Proposition 4.3 (What descends). *If two latent descriptions have the same intervention-indexed record laws for all a and t, they have the same τ_{obs}, and the same t_ε for every declared task and prior. A latent decomposition into source, path, wall and detector latencies need not descend. Only the combinations that the joint experiment identifies do. For example, a mean depending on $z = b + a\tau$ cannot separate τ from b, at any number of repetitions.*

5 The fold: snap and split

Theorem 5.1 ($\mathbb{Z}_2$-equivariant channels). *Let $\iota : b \mapsto 2d - b$, and let the observation law satisfy $P_{\iota(b)} = P_b$, as the overlap channel does (WT02).*

(a) *The exact quotient is $(x, r, |b - d|)$: an orbifold with fixed set the fold $b = d$.*

(b) *Every smooth invariant observable has zero normal derivative at the fold. The Fisher rank therefore drops there: the fold is an isotropy wall.*

(c) *Near the fold, with $o = o_0 - \kappa(b-d)^2 + O((b-d)^4)$, the identification modulus is $\omega(\varepsilon) \sim \varepsilon^{1/2}$. The Łojasiewicz exponent is 2, and the RLCT of the normal coordinate is $\frac{1}{4}$.*

(d) *An* anti-invariant *channel, $s \circ \iota = -s$, with nonzero normal derivative, such as the signed lead/lag channel $\tanh((b - d)/w_0)$, separates every pair and restores full rank. It* splits *the fold.*

Proof. (a) Identify the orbits. (b) Differentiate $o(\iota b) = o(b)$ at the fixed point: $-o'(d) = o'(d)$, so $o'(d) = 0$. (c) Invert the quadratic: $|b - d| \sim (\Delta o/\kappa)^{1/2}$. The RLCT of u^4 is $\frac{1}{4}$ (Rooting Closure). (d) Here $s'(d) \neq 0$, and s takes opposite values on the two members of each pair. Witness: X5, exact for the Gaussian overlap and tanh channels. □

Reading.

- A *snap* is passage through the orbifold point, where information about the rhythm collapses to second order.

- A *split* is the new anti-invariant protocol.

More data on the invariant channel never splits the fold. This is the structural form of "saturation needs a new protocol, not more data".

6 Timing eigenstructure: how uncertainty enters

Theorem 6.1 (Symmetries are exact null eigenvectors)**.** *If the record law is invariant under a one-parameter group acting on parameters with generator X, then $F\,X = 0$.*

Proof. Invariance gives $X \log p_\theta = 0$, so the score along X vanishes, and $X^\top F X = \mathbb{E}[(X \cdot s)^2] = 0$. Since F is positive semidefinite, $FX = 0$. □

Corollary 6.2 (The wall-tap null directions)**.**

- Dilation: *the delays are invariant under $(x, c, L) \mapsto k(x, c, L)$, so (x, c, L) is a null vector (X6).*
- Gain–contrast: *the amplitudes depend on gr, so $(r, -g)$ is a null vector.*

The anchors A_L, A_c and A_g are exactly the declarations that break these groups. This is why they, and only they, saturate the rank (WT01, WT08, WT09).

Proposition 6.3 (Wall eigenvalues)**.** *Channels whose means are linear in the wall contrast ε, with fixed covariance, give efficient information $\propto \varepsilon^2$ and a Cramér–Rao bound $\propto \varepsilon^{-2}$. This matches WT05–WT07, with slope* 2.000*.*

Theorem 6.4 (Timing precision; Woodward [5])**.** *For a pulse $s(t - \tau)$ in white noise of level σ^2, the timing information is*

$$F_\tau = \frac{E\beta^2}{\sigma^2}, \qquad \beta^2 = \frac{\int \dot{s}^2}{\int s^2}.$$

For a Gaussian pulse of width w, $\beta^2 = 1/(2w^2)$, which is verified symbolically (X7).

How uncertainty influences delay. Uncertainty enters only through these eigenvalues:

- energy times mean-square bandwidth over noise, for timing;
- ε^2, for the wall;
- zero, along symmetry generators.

It never adds a propagation time. A nuisance aggregate such as $z = b + a\tau$ creates an exact null that repetition cannot remove.

7 The rhythm transition

Theorem 7.1 (Dilation covariance)**.** *Let $\chi_{\mathrm{loc}}(T; h)$ be the ratio of dominant to total energy per period for taps of period T through a room response h. Then $\chi_{\mathrm{loc}}(kT; h(\cdot/k)) = \chi_{\mathrm{loc}}(T; h)$. If χ_{loc} is increasing in T, the crossing time at level θ is unique, and*

$$T_c[h(\cdot/k)] = k\,T_c[h].$$

Proof. Substitute $t = ks$ in every energy integral. Both energies scale by k, so their ratio is invariant. Uniqueness follows from monotonicity. □

Proposition 7.2 (Closed form for an exponential tail)**.** *Let the direct pulse have power e_0 and width τ_p, and the diffuse tail have energy density $a^2 e^{-2t/\tau_d}$. The tail energy from all earlier taps inside one period is $\frac{a^2\tau_d}{2} e^{-2T/\tau_d}$, by a geometric sum. Hence*

$$\chi_{\mathrm{loc}}(T) = \frac{e_0\tau_p}{e_0\tau_p + \frac{a^2\tau_d}{2}e^{-2T/\tau_d}}, \qquad T_c = \frac{\tau_d}{2}\ln\Big[\frac{\theta}{1-\theta}\cdot\frac{a^2\tau_d}{2e_0\tau_p}\Big].$$

This is verified symbolically, including covariance under $(T, \tau_d, \tau_p) \mapsto k(T, \tau_d, \tau_p)$ (X8).

Reading. The transition time is a room functional: about half the diffuse decay time, times a logarithm of a contrast ratio. The 35.58 ms of the original battery is the value of this functional for its declared synthetic room. It is not a universal constant. Any agreement with perceptual thresholds of a few tens of milliseconds is a separate, measured outcome. It is not implied by the acoustic theorem.

8 Ledger: structural versus empirical

Claim	Status	Basis
No distinguishability before mediated arrival	THEOREM	Theorem 2.2
Optimal error $\frac{1}{2}$ before arrival; monotone afterwards	THEOREM	Theorem 2.4
Delay structure is a Lawvere metric; seconds are not fixed by mediation	THEOREM	Proposition 2.3
First arrival is acoustic geodesic distance (Randers in wind)	THEOREM (imported core)	Corollary 3.2
Onset fragile; threshold time stable iff strict crossing	THEOREM	Section 4
Fold is an orbifold point; $\varepsilon^{1/2}$ modulus; signed channel splits	THEOREM	Theorem 5.1
Symmetry generators are null eigenvectors; anchors break them	THEOREM	Theorem 6.1
Timing precision $E\beta^2/\sigma^2$; wall information $\propto \varepsilon^2$	THEOREM	Theorem 6.4, Proposition 6.3
T_c formula and dilation covariance	THEOREM (model class)	Theorem 7.1, Proposition 7.2
Numerical values (8.6657 ms, 35.58 ms, 8.08 μs)	EMPIRICAL / ROOM CONSTANT	synthetic batteries
Human perception, memory, apparatus latencies	MEASURED, SEPARATE	not implied

9 Witnesses and limits

- `rg_walltap_lock.py` runs X1–X8, exact or symbolic, with no random sampling. Reruns are byte-identical.
  - X1–X2: the exact network and testing errors.
  - X3: onset fragility and flat crossing.
  - X4: eikonal timing with wind, and the image source.

  - X5: the fold and the signed split.
  - X6: the null eigenvectors.
  - X7: Woodward's formula.
  - X8: the T_c closed form and its covariance.

- *Limits.*
  - Finite propagation, Fermat/eikonal and Randers geometry are imported.
  - The T_c closed form is for an exponential diffuse tail; covariance holds for any response.
  - Hard moving fronts can produce nonregular models, where Cramér–Rao need not apply.
  - The wall-tap batteries remain the calibration of the protocol, not evidence for the theorems.

References

[1] F. W. Lawvere, Metric spaces, generalized logic, and closed categories, *Rend. Sem. Mat. Fis. Milano* 43 (1973) 135.

[2] R. Courant, D. Hilbert, *Methods of Mathematical Physics*, vol. II, Interscience, 1962.

[3] F. G. Friedlander, *The Wave Equation on a Curved Space-Time*, Cambridge, 1975.

[4] G. W. Gibbons, C. A. R. Herdeiro, C. M. Warnick, M. C. Werner, Stationary metrics and optical Zermelo–Randers–Finsler geometry, *Phys. Rev. D* 79 (2009) 044022.

[5] P. M. Woodward, *Probability and Information Theory, with Applications to Radar*, Pergamon, 1953.

[6] M. Rada, *From a Child's Wall-Tap Observation to Resolution Geometry*; *Wall-Tap Delay: Global*; *From the Wall to the Kernel*; *Experiments, Simulation and Descent*; *Rooting Closure*, 2026.